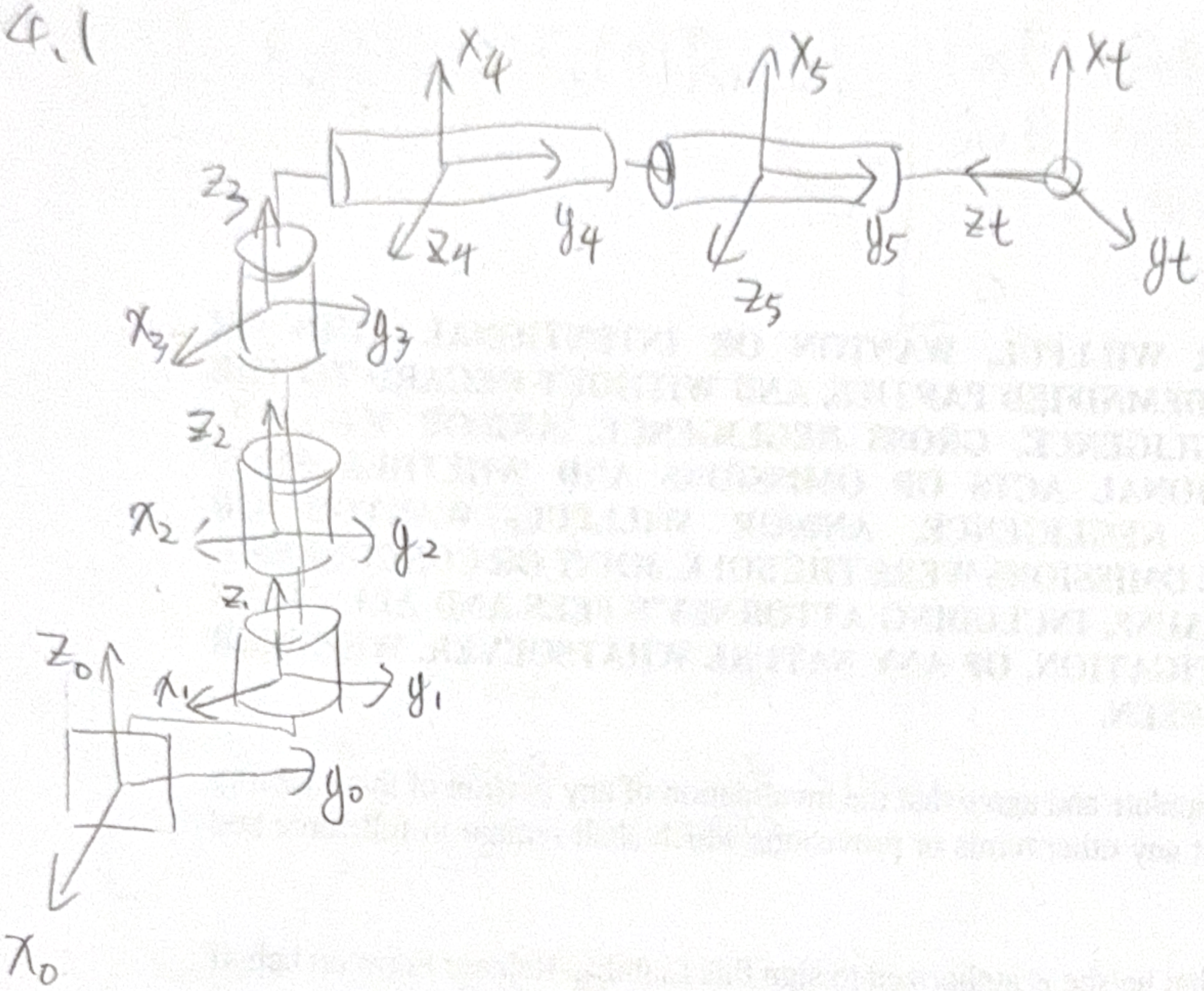
4.1



4.2

	a	α	d	θ
T_0	0.0388	0	0.0624	0
T_1	0.0304	90°	0.0542	q_1
T_2	0.1126	0	0.0280	q_2
T_3	0.1349	0	0	q_3
T_4	0.0611	90°	0	q_4
T_5	0	0	0.1004	q_5
T_t	0	0	0	0

9.3 $T = R_z(\theta)T_z(d)T_x(a)R_x(\alpha) = \begin{bmatrix} \cos\theta & -\sin\theta\cos\alpha & \sin\theta\sin\alpha & a\cos\theta \\ \sin\theta & \cos\theta\cos\alpha & -\cos\theta\sin\alpha & a\sin\theta \\ 0 & \sin\alpha & \cos\alpha & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$T_0 = \begin{bmatrix} 1 & 0 & 0 & 0.0388 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0.0624 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$T_1 = \begin{bmatrix} \cos q_1 & 0 & \sin q_1 & 0.0304\cos q_1 \\ \sin q_1 & 0 & -\cos q_1 & 0.0304\sin q_1 \\ 0 & 1 & 0 & 0.0542 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$T_2 = \begin{bmatrix} \cos q_2 & -\sin q_2 & 0 & 0.1126\cos q_2 \\ \sin q_2 & \cos q_2 & 0 & 0.1126\sin q_2 \\ 0 & 0 & 1 & 0.0280 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$T_3 = \begin{bmatrix} \cos q_3 & -\sin q_3 & 0 & 0.1349\cos q_3 \\ \sin q_3 & \cos q_3 & 0 & 0.1349\sin q_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$T_4 = \begin{bmatrix} \cos q_4 & 0 & \sin q_4 & 0.0611\cos q_4 \\ \sin q_4 & 0 & -\cos q_4 & 0.0611\sin q_4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$T_5 = \begin{bmatrix} \cos q_5 & -\sin q_5 & 0 & 0 \\ \sin q_5 & \cos q_5 & 0 & 0 \\ 0 & 0 & 1 & 0.1004 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$T_t = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

4.4

$T_B^I = T_0 T_1 T_2 T_3 T_4 T_5 T_t$